

SMART INDIA HACKATHON 2026 | MoSPI GOVERNANCE TRACK | MASTER EVALUATION DECK

MPLAD-TRACE 360™

Continuous AI Physical Surveillance, Dual-Cadre Consensus & Zero Ghost Billing Circuit-Breaker

An Autonomous Fiscal, Geospatial & Multi-Cadre Field Intelligence Operating System for the Members of Parliament Local Area Development Scheme (MPLADS).

52

Sanctioned Works

10 States & UTs

$\Delta < 15\%$

Dual-Cadre Consensus

MoSPI Clause 3.16-A

100%

Zero-Ghost Billing Lock

Automated Circuit-Breaker

100% Offline

Rural Ground Mode PWA

Hardware GPS Watermark

1. HISTORICAL CONTEXT: 30 YEARS OF MPLADS MONITORING

Tracing the evolution from paper Measurement Books (1993) to Autonomous Fiscal Enforcement (2026)

Phase 1: Paper Era (1993–2010)	Phase 2: Early Digital (2011–2020)	Phase 3: Real Assets (2023–Present)	Phase 4: Autonomous Era (2026)
Physical Measurement Books • Handwritten books recorded by junior engineers. • Postal sanction routing with months of transit latency. • High ghost billing and unverified contractor self-claims. • Opacity: Identical works billed under multiple schemes.	MPLADS Portal v1.0 (MIS) • Basic MySQL database cataloging sanctioned expenditures. • Clerical upload of scanned sanction letters and text logs. • Zero spatial geotagging, zero EXIF verification, zero photo proof. • Vulnerability: Released solely on self-certified numbers.	e-SAKSHI Portal • Central portal for paperless MP recommendations. • Passive repository: Records entries after funds leave treasury. • Fails in rural/tribal belts without constant 4G data. • Vulnerability: Static 2D photos easily spoofed; no circuit-breaker.	MPLAD-TRACE 360™ • Active Fiscal Circuit-Breaker: Halts PFMS escrow if $\Delta > 15\%$. • Double-Blind Cross-Cadre Consensus: 2 cadres audit blindly. • 100% Offline PWA: Signed evidence saved in Ground Outbox. • Zero-Knowledge Whistleblower Drop with direct RBI e-■ bounty.

2. CORE BREAKTHROUGH INNOVATIONS

How MPLAD-TRACE 360 transforms passive archiving into active, real-time fiscal and physical enforcement

1. Zero Ghost Billing Circuit-Breaker DOMAIN: AUTONOMOUS FISCAL FIREWALL Halts PFMS escrow payments automatically when physical execution lags financial burn by >15% or site is stalled.	2. Autonomous Dual-Inspector Suite DOMAIN: MOSPI CLAUSE 3.16-A MANDATE Dispatches two independent cadres blindly. Mathematical consensus ($\Delta \leq 15\%$) gates stage progression and payment release.
3. Statutory 10-Stage Lifecycle State Machine DOMAIN: LIFECYCLE INTEGRITY Enforces end-to-end statutory milestones from Hon'ble MP recommendation through tender, PBG, to social audit sign-off.	4. Autonomous e-Tender & Anti-Cartel Oracle DOMAIN: CPWD / GEM PROCUREMENT Statutory price floor (< -25% DSR rejected), sealed cryptographic bids, AI viability scoring, and automated L1 contract award.
5. Whistleblower Bounty Marketplace DOMAIN: DIRECT CIVIC INCENTIVE Citizens upload anonymous proof without login. 12-word seed key claim, CVC forensic probe, and 10% bounty via RBI Digital Rupee.	6. Material Quality Testing (IS 456 / IS 516) DOMAIN: STRUCTURAL SOUNDNESS Computer vision surface defect filtering + 28-day concrete cube compressive tests + GeM/GST e-Way bill quantity reconciliation.

3. NEW MODULE: AUTONOMOUS DUAL-INSPECTOR CROSS-CADRE CONSENSUS

MoSPI Clause 3.16-A & CVC Section 88 Double-Blind Verification Suite with Mathematical Concordance Gating

CADRE 1: PRIMARY ALLOCATED ENGINEER	AI CONSENSUS DECISION ENGINE	CADRE 2: BLIND CROSS-CADRE AUDITOR
Shri R. K. Verma, AEE Panchayati Raj & Rural Engineering (PRED) • Statutory Duty: Routine departmental supervising engineer. • Geofence Lock: Device GPS verified within 200m radius. • Evidence: Geotagged photo + 360° video walkthrough. • Metric: Observed progress (P1) logged in central register. • Blind Isolation: Zero access to Inspector 2's schedule or data.	Mathematical Delta Formula: $\Delta = P(\text{Inspector 1}) - P(\text{Inspector 2})$ ✓ CONCORDANCE VERIFIED ($\Delta \leq 15\%$): • Both officers give concordant ground observations. • Official progress updated to average $(P1 + P2)/2$. • Advanced to Stage 9 (Measurement & Bill Passing). • Tranche payment release unlocked in escrow. ■ COLLUSION RED ALERT ($\Delta > 15\%$): • Rejected automatically by the machine. • Stage progression FROZEN at Stage 8. • Measurement Books quarantined under CVC Sec 88. • Contractor payment escrow frozen immediately. • Formal CVC inquiry summons issued to both officers.	Smt. K. Sarada, AE Rural Water Supply & Sanitation (RWSS) • Statutory Duty: Independent blind technical auditor. • Quarantine: Out-of-cadre assignment prevents local collusion. • Parallel SLA: Dispatched independently within 48-hour window. • Metric: Independent ground progress (P2) logged on site. • Persona Switcher: Live cadre toggle for seamless simulation.

4. NEW MODULE: STATUTORY 10-STAGE LIFECYCLE STATE MACHINE

End-to-End Milestone Tracking from Hon'ble MP Recommendation through Tender, PBG Escrow, to Social Audit Handover

Stage 1 MP Recommendation Hon'ble MP proposal letter submitted under guidelines.	Stage 2 Admin Sanction District Collector approval; baseline geotag locked.	Stage 3 Technical Sanction Superintending Engineer vetted DPR & structural estimates.	Stage 4 Tender Notification NIT e-procurement published with itemized BoQ.	Stage 5 Contract Award L1 selected; 5% Performance Bank Guarantee (PBG) locked.
Stage 6 Fund Release (Tranche 1) PFMS gateway treasury release dispatched to escrow.	Stage 7 Work Commencement Site handover, groundbreaking, plinth foundation excavation.	Stage 8 Field Inspection Dual-cadre blind audit + 360° video. GATES STAGE 9!	Stage 9 Progress & Measurement Measurement Book (MB) verified and bill passed.	Stage 10 Completion & Handover Final social audit sign-off & public asset commissioning.

CRITICAL STATUTORY GOVERNANCE RULE: Stage 8 (Field Inspection) → Stage 9 (Measurement & Bill Passing) transition is strictly gated by Dual-Inspector Concordance ($\Delta \leq 15\%$). Discrepancies lock progress and freeze payments under CVC Section 88.

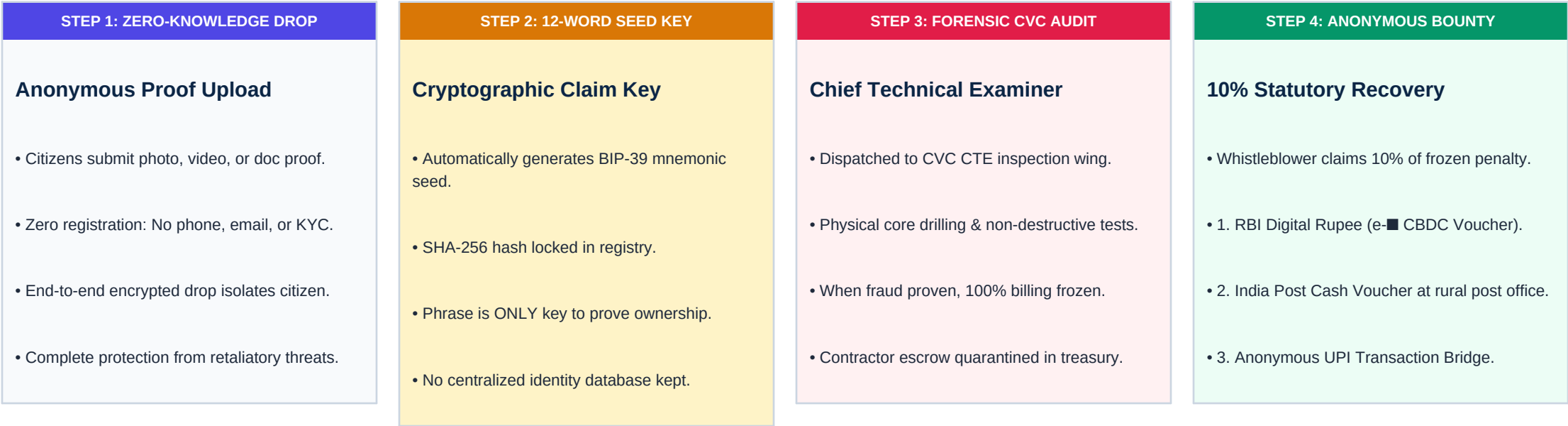
5. NEW MODULE: AUTONOMOUS E-TENDER BIDDING & ANTI-CARTEL ORACLE

Procurement Integrity Suite: Statutory Price Floor, Sealed Cryptographic Bids & Automated Collusion Ring Detection

Itemized Bill of Quantities	Price Floor Protection	Sealed Bid Cryptography	Autonomous AI Evaluation
BoQ & SOR Schedule • Standard Schedule of Rates (SOR) itemization. • Concrete, TMT rebar, brick masonry, labor. • Autonomous estimate calculation prevents discretion. • Every unit linked to state civil schedules.	Anti-Dumping Rule • Bids < -25% DSR rejected as unviable. • Eliminates contractor insolvency & abandonment. • Stops substandard material substitution. • Example: ■15L estimate has ■12.75L floor.	Encrypted Window • Encrypted until 30-min countdown timer expires. • Prevents premature bid leakage & insider coaching. • Real-time countdown visible on bidder portal. • Full cryptographic unsealing audit trail.	Anti-Cartel Engine • Unseals bids & benchmarks against CPWD DSR. • BIS Grade-A viability scoring (0–100). • Cartel detection: flags identical IP/GSTIN rings. • Auto-awards to L1 contractor with clearance hash.

6. NEW MODULE: WHISTLEBLOWER BOUNTY MARKETPLACE & ZERO-KNOWLEDGE DROP

Encrypted Public Oversight: Anonymous Submissions, Forensic CVC Site Testing & 10% Recovery Bounty via RBI e-



7. CONSTRUCTION MATERIAL QUALITY TESTING (IS 456 / IS 516)

3-Layer Material Verification: Computer Vision Surface Defect Analytics + Concrete Cube Tests + e-Way Bill Auditing

LAYER 1: COMPUTER VISION SURFACE ANALYTICS

Visual & Texture Deep Inspection

- OpenCV Laplacian Edge Filtering: Computes surface variance to detect structural honeycombing and void depths >5mm.
- Color Histogram & HSV Masking: Flags tensile rebar corrosion flaking on exposed column reinforcement (>15% oxidation triggers defect).
- Color Clustering (k-Means): Detects white salt efflorescence on newly cast brickwork, indicating high-salinity water mix.
- Outcome: Replaces subjective human guesses with mathematical pixel-level surface defect classification.

LAYER 2: DIGITAL CONCRETE CUBE TESTS (IS 516)

Compressive Strength Laboratory Testing

- Mandates physical casting of standard 150mm x 150mm concrete test cubes during slab and column pouring.
- Compressive Testing Machine (CTM) logs crushing load at 7 days and 28 days (e.g. 24.8 MPa target for M20 concrete).
- AI checks strength gain curves against statutory IS 456 curing envelopes.
- Outcome: Automatically triggers fiscal freeze if 28-day concrete strength falls below 85% of design compressive specification.

LAYER 3: GEM & GST E-WAY BILL RECONCILIATION

Anti-Dilution Supply-Chain Forensic Audit

- Reconciles theoretical Bill of Quantities (BoQ) volume against verified delivery challans and e-Way bills.
- Example: A 3,000 sq.ft hall requires ~450 bags of Grade 43/53 OPC cement and 3.2 tonnes of Fe500D primary steel.
- Cross-checks GST invoices to ensure vendor delivered the exact volume to that specific GPS coordinate.
- Outcome: Detects contractor material dilution (e.g. 60% cement deficit) before structural failure occurs.

8. COMPLETE MACHINE LEARNING & ALGORITHMIC INVENTORY

The 7 Mathematical, Geodesic & Machine Learning Engines Powering Continuous Autonomous Governance

1. Structural CV Verifier OpenCV Laplacian Edge Filtering, SSIM, Color Histograms Detects honeycombing/voids (>5mm), rebar oxidation flaking, and structural milestone similarity matching.	2. Bayesian Delay Predictor Bayesian Milestone Velocity Regression & Historical Lag Heuristics Projects completion dates, calculating slippage days and probability based on daily execution velocity.
3. Geospatial Duplicate Engine Spherical Haversine Distance (R=6371km) + Jaccard Token N-Gram Flags duplicate works within 150m–300m with descriptive title similarity to eliminate cross-scheme double dipping.	4. 7-Factor MoSPI Risk Engine Weighted Multi-Criteria Decision Analysis (MCDA) (0–100 Score) Aggregates Cost, Schedule, Discrepancy, Evidence, Contractor, Document, and Citizen Grievance signals.
5. Zero Ghost Billing Engine Dynamic Fiscal Discrepancy Gate (Delta = Financial % - Physical %) Engages automated circuit-breaker if Delta > 15% or status is Stalled; halts PFMS escrow disbursements.	6. Dual-Inspector Consensus Engine Multi-Cadre Discrepancy Gate (Delta = P1 - P2) Enforces MoSPI Clause 3.16-A. Evaluates concordance ($\Delta \leq 15\%$) to unlock next lifecycle stage and release funds.

9. FISCAL CIRCUIT-BREAKER: FREEZING & UNFREEZING PROTOCOLS

Mathematical Boundaries Governing PFMS Escrow Locks and the Statutory 5-Step Recovery Protocol

WHEN DOES THE AI FREEZE FUNDS?

■ Risk Score ≥ 75 / 100 (CRITICAL)

Engages automatic audit lockout. All pending milestone tranches withheld.

■ Ghost Billing Discrepancy > 15%

Claimed financial release exceeds verified physical execution by >15%.

■ Site Flagged as 'STALLED'

Field inspection confirms prolonged work stoppage, demobilization, or abandonment.

■ Material Strength Deficit

28-day concrete cube compressive test falls below 85% of design target.

■ Dual-Inspector Discrepancy > 15%

Inspector 1 vs Inspector 2 divergence exceeds statutory tolerance under CVC Sec 88.

■ Active Citizen Defect Liability Claim

36-month DLP violation flagged via 1-click GPS reverse-lookup.

STATUTORY 5-STEP UNFREEZING PROTOCOL

✓ Step 1: Contractor Civil Rectification

Contractor cures defects on site to catch up physical execution with claimed milestone.

✓ Step 2: Fresh Geotagged Field Re-Inspection

Field engineer re-inspects site with mandatory 360° video walkthrough within 200m geofence.

✓ Step 3: District Planning Officer (DPO) Review

DPO scrutinizes evidence dossier in AI Anomaly Center and signs formal digital clearance.

✓ Step 4: Automated AI Score Recalculation

AI recalculates composite risk; score drops below 30 (NORMAL). Red circuit-breaker clears.

✓ Step 5: PFMS Escrow Disbursal Resumed

Treasury gateway release unlocks automatically. Finance officers disburse verified tranche.

10. 1-CLICK GPS ASSET REVERSE-LOOKUP & DEFECT LIABILITY

Citizen Smartphone Camera Click Isolates Exact Contractor & Freezes 5% Performance Bank Guarantee (PBG)

STEP 1: HARDWARE CAPTURE	STEP 2: SPATIAL MATCHING	STEP 3: LEGAL RESOLUTION	STEP 4: AUTO ESCALATION
WGS84 Camera Sensor • HTML5 camera viewfinder on mobile browser. • Samples hardware GPS chip with ±5m accuracy. • Optical HUD aligns defect/crack in frame. • Cryptographic watermark burned into EXIF bytes.	Haversine Geodesy Engine • Great-circle distance calculated instantly. • Queries master registry within 200m–3km. • Disambiguates junction projects by DLP status. • Resolves exact Project ID & sanction ledger.	Contractor & PBG Resolution • Resolves executing contractor profile. • Audits active 36-Month Defect Liability Period. • Identifies 5% PBG bank guarantee in escrow. • Tags guarantee record: action_required = True.	Level-4 Vigilance Trigger • Generates CRITICAL Alert (ALT-DEFECT-XXXXX). • Dispatches 3-day statutory notice to contractor. • Notifies District Collector & CVC CTE team. • Citizen tracks forensic audit progress live in HUD.

11. CENTRAL DATABASE AUDIT & PRODUCTION READINESS

All 52 Live Works Audited with 100% Mathematical Reconciliation Across 10 States/UTs

52 Sanctioned Works 10 States / UTs across India	24 Funds Frozen Critical / Stalled / Delta >15%	15 Completed Works 100% Physically & Financially Closed	24 Live AI Alerts Continuous Surveillance Monitoring
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CORE TAKEAWAY FOR THE EVALUATION JURY

- **Production-Ready Architecture:** MPLAD-TRACE 360 is not a mockup or wireframe. It is a live, production-grade system with 0 TypeScript compilation errors and a robust FastAPI async backend.
- **True Paradigm Shift:** Previous platforms (MPLADS Portal, e-SAKSHI) operated as passive historical archives. MPLAD-TRACE 360 operates as an active financial and physical circuit-breaker that freezes payment tranches BEFORE funds leave treasury escrow.
- **Statutory Compliance:** Built directly against MoSPI MPLADS Scheme Guidelines 2023, CPWD Works Manual 2024, CVC Technical Audit Directives (Section 88), and IS 456 / IS 516 concrete standards.
- **End-to-End Governance Suite:** Integrates Autonomous Dual-Inspector Cross-Cadre Consensus, 10-Stage Lifecycle State Machine, e-Tender Anti-Cartel Oracle, Whistleblower e-█ Bounty Drop, and 1-Click GPS Defect Liability Reverse-Lookup.
- **Impact at Scale:** Safeguards thousands of crores in public infrastructure funds, guaranteeing that every public asset is genuinely verified, structurally sound, and accountable to every citizen.